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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 57.4

$$I = \int \frac{1}{\tan x - 1} dx$$

$$\text{Stel } t = \tan x \Rightarrow dx = \frac{1}{1+t^2} dt$$

Dus:

$$I = \int \frac{1}{t-1} \cdot \frac{1}{1+t^2} dt = \int \frac{1}{(t-1)(t^2+1)} dt$$

$$\frac{1}{(t-1)(t^2+1)} = \frac{A}{t-1} + \frac{Bt+C}{t^2+1}$$

$$A = \frac{1}{t^2+1} \Big|_{t=1} = \frac{1}{2}$$

$$Bi + C = \frac{1}{t-1} \Big|_{t=i} = \frac{1}{i-1}$$

$$= \frac{-1-i}{(-1+i)(-1-i)} = \frac{-1-i}{2}$$

$$= -\frac{1}{2} - \frac{1}{2}i$$

$$B = -\frac{1}{2}, C = -\frac{1}{2}$$

$$I = \frac{1}{2} \int \frac{1}{t-1} dt - \frac{1}{2} \int \frac{t+1}{t^2+1} dt$$

$$= \frac{1}{2} \int \frac{1}{t-1} dt - \frac{1}{4} \int \frac{2t}{t^2+1} dt - \frac{1}{2} \int \frac{1}{t^2+1} dt$$

$$= \frac{1}{2} \ln|t-1| - \frac{1}{4} \ln(t^2+1) - \frac{1}{2} \text{Bgtan } t + c$$

Terug substitueren:

$$I = \frac{1}{2} \ln|\tan x - 1| - \frac{1}{4} \ln(\tan^2 x + 1) - \frac{1}{2} x + c$$

$$1 + \tan^2 x = \frac{1}{\cos^2 x}$$

Dus:

$$I = \frac{1}{2} \ln \left| \frac{\sin x - \cos x}{\cos x} \right| - \frac{1}{4} \ln \left(\frac{1}{\cos^2 x} \right) - \frac{1}{2} x + c$$

$$= \frac{1}{2} \ln|\sin x - \cos x| - \frac{1}{2} x + c$$

References